

TELCO CUSTOMER CHURN ANALYSIS REPOT

BUSINESS ANALYTICS CASE STUDY – WEEK 5

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1. Introduction

Customer retention is an important challenge for companies operating in the telecommunications sector. When a customer leaves the company, it can result in revenue loss and require additional investment to acquire a replacement customer.

As part of my Week 5 at AnalystLab Africa, I sought to understand why some customers leave the company and which are the main factors associated with churn.

To address this issue, I analyzed a telecommunications customer dataset and developed an interactive Power BI dashboard to visualize key indicators, identify customer profiles most exposed to churn, and provide relevant business recommendations.

2. Business Problem Statement

The main question addressed in this study is :

Why are customers leaving the company, and which factors are most strongly associated with Customer Churn?

The objectives were therefore to :

measure the extent of customer churn ;

identify the customer categories most affected ;

analyze churn across different customer and service characteristics ;

identify the main factors associated with churn ;

translate the findings into practical recommendations to improve customer retention.

3. Dataset Description

The analysis was conducted using the Telco Customer Churn dataset, containing information on 7043 customers.

The dataset includes several categories of information that help analyze customer behavior, including:

demographic characteristics ;

customer tenure ;

contract type ;

monthly charges ;

payment method ;

Internet service ;
technical support ;
Online Security ;
Online Backup ;
Dependents ;
Senior Citizen status.

Before the analysis, the dataset was reviewed to verify data types, assess data consistency, and check for missing or inconsistent values.

4. Data Analysis Process

The analysis was mainly conducted using Microsoft Power BI.

The main steps included :

Data exploration – understanding the structure of the dataset and the available variables.

Data preparation – reviewing and standardizing data types and checking data quality.

KPI creation – creating key indicators to monitor total customers, churned customers, retained customers and churn rate.

Exploratory analysis – analyzing churn across contract type, tenure, monthly charges, payment method, Internet service, technical support and additional services.

Data visualization – developing an interactive dashboard with filters and visualizations.

Business analysis – identifying key churn drivers and customer segments.

Recommendations – translating the findings into actionable retention strategies.

The final dashboard is organized around four main areas :

Overview → Churn Drivers → Segments → Actions

5. Key Findings

The analysis highlighted several important findings.

Contract Type

Customers with Month-to-month contracts accounted for 1655 churned customers, compared with 166 for one-year contracts and 48 for two-year contracts.

This indicates that short-term contract customers represent the group most strongly associated with churn.

Internet Service

Fiber optic customers accounted for 1297 churned customers, compared with 459 for DSL and 113 for customers without Internet service.

This suggests that the perceived value, pricing and customer experience associated with Fiber optic services deserve further attention.

Payment Method

The Electronic check payment method was associated with 1071 churned customers, the highest number among the payment methods analyzed.

Technical Support

Customers without Tech Support accounted for 1446 churned customers, compared with 310 among customers who had technical support.

This highlights the potential importance of technical assistance in customer retention.

Online Security and Online Backup

The analysis also showed higher churn counts among customers who did not subscribe to certain additional services, particularly Online Security and Online Backup.

These services may therefore represent potential retention levers.

6. Business Insights

The results suggest that Customer Churn is not associated with a single factor. Several characteristics related to contracts, tenure, services and customer experience are associated with different levels of churn.

Contract type stands out as an important factor in the analysis. Month-to-month customers show a substantially higher churn volume than customers with longer-term contracts.

Early customer engagement is another important area of attention. Customers with shorter tenure may require closer follow-up to identify dissatisfaction and potential churn risks at an early stage.

The results related to Tech Support, Online Security and Online Backup also suggest that additional services may help strengthen the perceived value of the company's offering and potentially support customer retention.

Finally, the high churn levels observed among Fiber optic customers and Electronic check users indicate segments that deserve further investigation and targeted actions.

Note: These results represent associations observed in the dataset. They do not, by themselves, establish that a specific factor directly causes customer churn.

7. Recommendations

Based on the findings, the following recommendations are proposed:

- Encourage Long-Term Contracts

Introduce loyalty benefits, preferential pricing or other incentives to encourage Month-to-month customers to move to one-year or two-year contracts.

- Strengthen Early Customer Engagement

Implement proactive onboarding and follow-up during the first months of the customer relationship to identify dissatisfaction early and reduce avoidable churn.

3. Improve the Perceived Value of Internet Services

Review service quality, pricing and customer experience, particularly for Fiber optic customers, and develop targeted offers where appropriate.

- Promote Value-Added Services

Promote Tech Support, Online Security and Online Backup as potential retention tools by clearly communicating their benefits to relevant high-risk customers.

- Develop Targeted Retention Campaigns

Combine information about contract type, tenure, monthly charges and subscribed services to identify high-risk customers and provide personalized retention actions rather than applying a one-size-fits-all approach.

8. Conclusion

This Business Analytics Case Study demonstrates how customer data can be transformed into actionable business insights.

Using Power BI, the analysis identified several customer groups and service characteristics associated with higher churn levels, including Month-to-month contracts, Fiber optic services, Electronic check payments and the absence of Tech Support.

The findings highlight the importance of long-term contract adoption, early customer engagement, service value and targeted retention strategies.

This project also allowed me to strengthen my practical skills in Business Analytics, KPI Analysis, Data Visualization and Data Storytelling, while developing my ability to transform data into actionable insights and decision-oriented recommendations.